

CCNA 200-301 Study Guide

Chapter 31 — VLANs Part 1

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1. Chapter Overview

Introduction to Virtual Local Area Networks (VLANs), broadcast-domain segmentation, VLAN benefits, VLAN creation, access-port assignment, and trunk ports between switches.

2. What is a VLAN?

VLAN = Virtual Local Area Network. A VLAN logically divides a switch network into separate broadcast domains. Each VLAN represents an independent broadcast domain.

3. Why VLANs?

A flat network can place many devices in one broadcast domain. As broadcast traffic grows, performance can be affected. VLANs create smaller, independent broadcast domains.

Benefit	Purpose
Security	Helps logically isolate groups of devices and limit broadcast scope.
Network Performance	Reduces the size of broadcast domains and broadcast traffic.
Management	Makes network resources and device groups easier to organize and manage.

4. Create and Name a VLAN

Enter global configuration mode, create the VLAN, then give it a name.

```
configure terminal
vlan 10
name SALES
```

5. Verify VLANs

```
show vlan
do show vlan
```

do show vlan can be used while inside configuration mode.

6. Assign a Port to a VLAN

Configure the port as an access port and assign it to the VLAN.

```
interface fa0/1
switchport mode access
switchport access vlan 10
```

7. Assign a Range of Ports

Apply the same access-VLAN configuration to multiple interfaces.

```
interface range fa0/1 - 4
switchport mode access
switchport access vlan 10
```

8. Trunk Ports

A trunk port is used between interconnected switches to carry traffic belonging to multiple VLANs.

```
interface fa0/24
switchport mode trunk
```

9. Practical Concept

If two switches contain VLAN 10, VLAN 20, and VLAN 30, their inter-switch link can be configured as a trunk so traffic from the multiple VLANs can cross that link.

10. Exam Notes

- **VLAN = Virtual Local Area Network.**
- **Each VLAN is a separate broadcast domain.**
- **Main benefits: Security, Network Performance, and Management.**
- **switchport mode access → configures an access port.**
- **switchport access vlan 10 → assigns the port to VLAN 10.**
- **interface range → applies configuration to multiple interfaces.**
- **switchport mode trunk → configures a trunk port.**
- **A trunk carries traffic from multiple VLANs between switches.**

11. Quick Revision

VLAN:

```
vlan 10  
name SALES
```

Access:

```
interface fa0/1  
switchport mode access  
switchport access vlan 10
```

Range:

```
interface range fa0/1 - 4  
switchport mode access  
switchport access vlan 10
```

Trunk:

```
interface fa0/24  
switchport mode trunk
```

Verify:

```
show vlan
```

Source scope: based on the supplied VLAN Part 1 video summary; no additional topics were added.